Reliability, Faithfulness, and the Limits of Post-hoc Explanations of Opaque Scientific Models

ICML International Conference On Machine Learning

PhilML Workshop

Nick Oh
socius labs
nick.sh.oh@socius.org

Helen Jin
University of Pennsylvania
helenjin@seas.upenn.edu

TL;DR

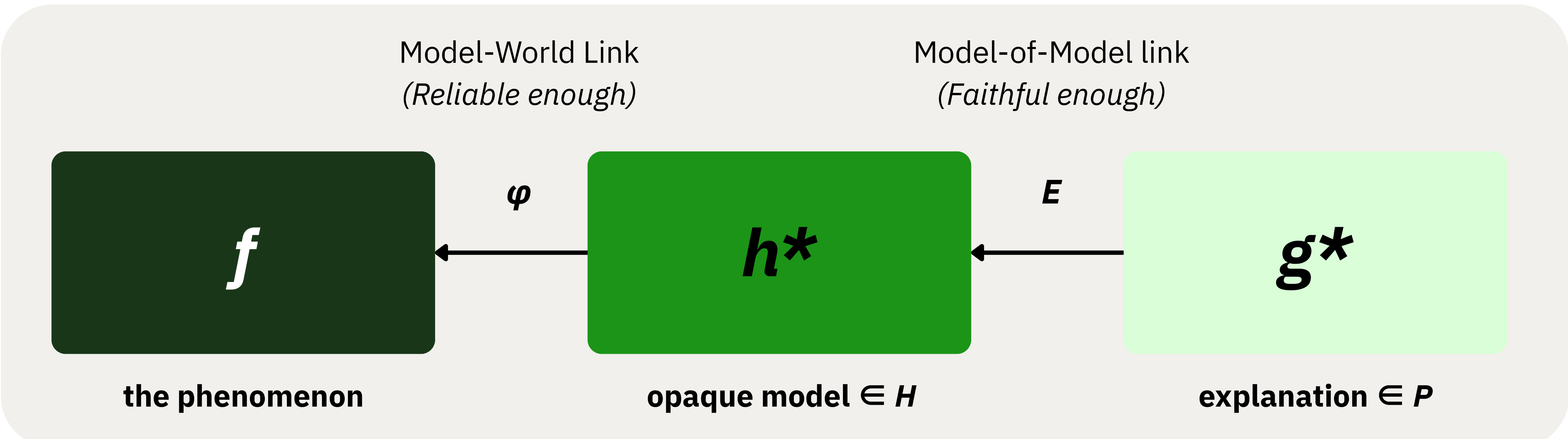
Explaining the model faithfully is *not* explaining the world, however reliable the model may be.

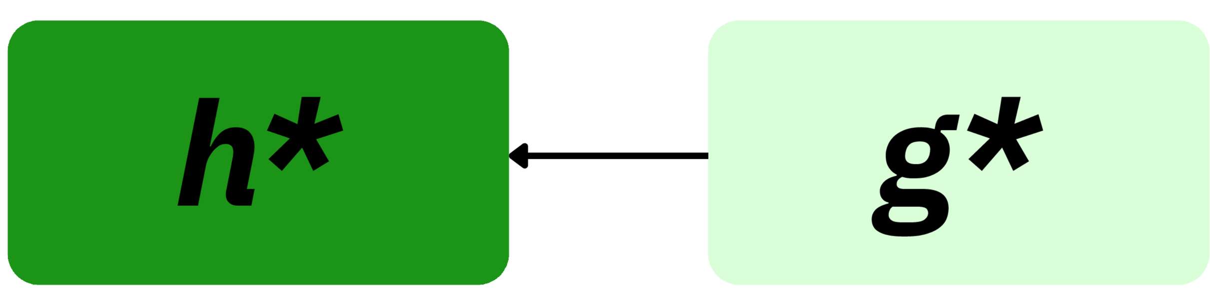
Faithfulness is about computation, reliability about prediction. Neither is about the world's structure.

f* : *X* → *Y denote the true function representing the natural phenomenon of interest.

A predictive model is trained to approximate *f* by selecting a hypothesis ***h* ∈ *H***, where *H* is a hypothesis space representable by the DNN.

For an opaque model *h*, a post-hoc explanation method *E* produces an explanatory representation ***g* = *E*(*h*) ∈ *P***, drawn from a space of possible explanations *P* (e.g., SHAP, saliency map).





The explanation ***g**** faithfully reports that ***h**** relies on some feature *j*.
The claim the scientist wants is about ***f***; that *j* belongs to the phenomenon.

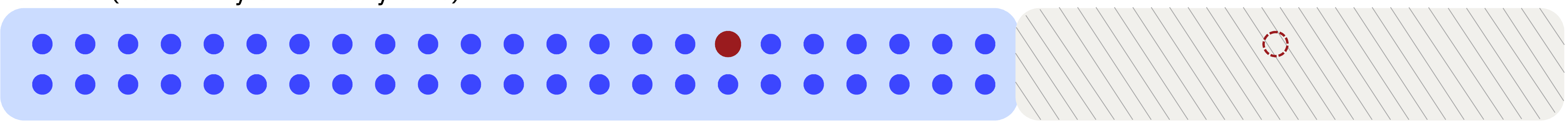
Accessible Observations

Obtained by sampling populations. Validation tests predictive reliability only on these observations.

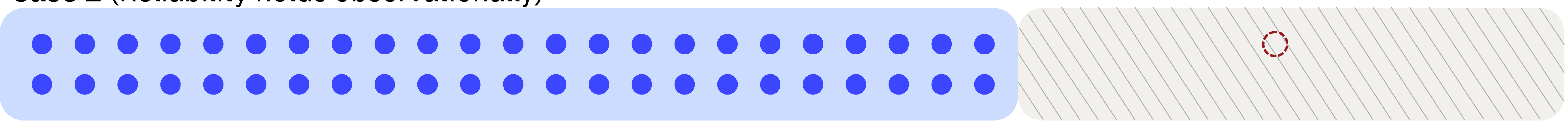
Dissociating Observations

Reachable only through intervention on the phenomenon.

Case 1 (Reliability eventually fails)



Case 2 (Reliability holds observationally)



Case 3 (Perfect reliability)